BBO capstone project presentation template

Develop a concise presentation that captures:

1. **An overview of your BBO approach**
   - Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
   - Briefly describe your overall strategy and how your process fits together.

Overall Approach

My approach has been heuristic and exploratory, reflecting the idiosyncratic nature of each function and the development of my technical skills. Over successive rounds, experimentation gradually led to more targeted, function-specific solutions.

1. Early Rounds: Learning GP and Bayesian Optimisation

- Heavy emphasis on Gaussian Processes (GPs) and Bayesian Optimisation (BO).
- Experimented with kernels, hyperparameter tuning and acquisition functions: EI, PI and UCB.
- Focused on understanding function behaviour, identifying promising regions and balancing exploration with exploitation.

2. Middle Rounds: Ensembles & Challenger Models

- As my skills developed, I introduced and experimented with (and at times ensembled) more sophisticated algorithms such as neural networks, TuRBO and HEBO. However, in one case (F8), I used a simple regression model and brute-force search to identify a candidate maximum.
- Compared competing model recommendations rather than relying exclusively on GP predictions.
- Introduced more systematic validation, such as LOOCV.
- Recognised that greater model complexity did not necessarily improve predictive accuracy or optimisation performance.

3. Late Rounds: Convergence and Simplification

- Progressively concentrated on models and techniques that held the most potential. This included retaining local GP refinement for selected lower-dimensional functions and TuRBO for F6 and F7.
- Adapted strategies to observation noise and model disagreement, which took judgement and often quite a bit of diagnostics.
- I believe that F5 and F8 are solved/maximised, which removed the need for further optimisation.
- Finally, I refactored Python notebooks. This involved further simplifying code, removing redundant diagnostics and simplifying data preparation in order to improve reproducibility.

4. Current Methodology

- **Process:** Data preparation → Model selection → Validation → Candidate generation → Model comparison → Query selection → Iteration.
- **Highly bespoke:** No single model or algorithm is universally superior. Effective optimisation requires selecting the appropriate tools for each function. This meant a large investment in time each week.

2. How your strategy has evolved

- Describe the key changes in your approach since your early rounds.
- What influenced these changes – data trends, feedback, model performance or intuition?
- Identify which principles or heuristics now guide your query decisions.

There is one important lesson: a more sophisticated model does not necessarily produce better results, particularly when developing new skills. My approach gradually shifted from experimenting with as many techniques as possible to being more selective about which techniques are useful and when to apply them.

Factors Influencing This Were:

1. Function Behaviour

- The eight functions behaved differently. Some responded well to local refinement, while others were difficult to model or produced inconsistent results.
- I became more sceptical of predictions suggesting substantial improvements outside of previously explored regions. Comparing predictions with actual results became increasingly important.

2. Model Evaluation

- As my skills improved, I introduced neural networks, ensembles and TuRBO to challenge the original GP approach. This, at times, worked well but often caused greater confusion and was time-consuming.
- I placed greater emphasis on LOOCV, predictive uncertainty and agreement between models. This was particularly true for F3 and F4.
- Comparing alternative approaches helped identify where additional complexity was useful and where it was not.

3. Refinement and Simplification

- As noted above, towards the later rounds, I began removing models, diagnostics and code that contributed little to query decisions.
- I increasingly concentrated on approaches that had demonstrated performance and shifted towards simpler, more reliable and reproducible analysis.

Key Lessons Learned: What I Would Do Differently

- **Start with the observations:** Examine the actual results (best-performing regions, trends and unexpected outcomes) before fitting models.
- **Think more carefully about the problem:** Consider factors such as dimensionality, noise, available observations and function behaviour. Only then select an appropriate approach rather than experimenting with every available technique.
- **Use challenger models selectively:** When different models produce conflicting predictions, use alternative models to investigate further. However, disagreement between acquisition functions is somewhat different as they balance exploration and exploitation.
- **Be more deliberate about exploration:** Particularly when the description notes potential multiple peaks.
- **Keep code simple and reproducible.** I've learned to "tidy as I go", meaning removing obsolete code and diagnostics and documenting decisions throughout the process. Leaving that towards the end created additional work.

3. Patterns, data and insights

- Highlight the most meaningful trends you've observed across your data points or model evaluations.
- Identify which variables or behaviours seem to influence results the most.
- Explain how these observations have shaped your understanding of the search process.

The eight functions produced quite different results. One interesting observation is that dimensionality alone doesn't explain optimisation difficulty. F7, with six variables, has been relatively responsive to local optimisation, while F3 and F4 have been more challenging despite having fewer variables. Equally, I found functions with more observations (F8) more straightforward once the right tool was selected.

Key Observations

Function	Key trends and influential variables	Implications for the search
F1	Improvements along a narrow trajectory near (0.67, 0.70), with diminishing gains. Variables seem interdependent.	Explore perpendicular directions vs repeatedly refining the same trajectory.
F2	R11 improved the maximum by 12%. x1 appears influential near 0.51, with x2 around 0.55.	Vary x1 independently to test local sensitivity and the extent of the promising region.
F3	Identical inputs had different outputs. GP predictive accuracy remained weak despite transformations.	Account for noise; prioritise robust predictions, model agreement and uncertainty.
F4	R11 fell from a previous maximum of 0.731 to 0.199. GP, NN and TuRBO disagree substantially.	Treat predicted improvements cautiously. Validate competing models before selecting queries.
F5	Outputs increased towards upper boundaries; 8662.48 repeatedly achieved at (1,1,1,1).	Boundary solutions can be optimal albeit not intuitive. Considered maximised unless another region has been missed.
F6	R11 achieved the first positive output. Improvement accompanied increasing x1–x3 and decreasing x4–x5.	Coordinated variable movements matter. Continue local exploration using TuRBO.
F7	Successive local improvements reached 3.283. TuRBO recommendations remained relatively stable.	Consistent local refinement appears effective; prioritise exploitation.
F8	Output of 10 repeatedly achieved with adjacent points tested.	High numerical precision is not always necessary. Likely maximised.

Summary

- Dimensionality alone does not determine difficulty (F3/F4 versus F7).
- Model disagreement and observation noise complicate optimisation.
- The search strategy depends on the evidence, e.g. targeted exploration for F1, continued exploitation for F7.

4. **Decision-making and iteration**
   - Explain how you balance exploration and exploitation.
   - Describe one or two examples of strategic decisions. What worked, what didn't, and why?
   - Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

Balancing Exploration and Exploitation

Initially, I placed considerable emphasis on exploration, partly because I had limited observations and partly because I was experimenting with different modelling techniques.

As the rounds progressed, I became more focused on exploiting regions that had produced promising results. However, I also recognised that repeatedly searching the same region could cause me to overlook other opportunities.

My decisions increasingly depended on function behaviour.

I also became less inclined to follow acquisition-function recommendations without questioning whether they made sense.

Examples of Strategic Decisions

Function	Decision and reasoning	Outcome or lesson
F1	After successive small improvements along a narrow trajectory, selected a perpendicular displacement for R12.	Tests whether the established search direction has overlooked useful information. A deliberate shift from exploitation to targeted exploration.
F4	Compared GP recommendations with neural networks and TuRBO after substantial disagreement and a poor R11 result.	Model disagreement revealed uncertainty beyond the GP's estimates. High predicted improvement does not "guarantee" a better outcome.
F6	Continued TuRBO after R11 produced the first positive output, rather than restarting a global search.	Exploits evidence of a productive local trajectory while maintaining exploration within the trust region.
F7	Continued local TuRBO refinement following successive improvements.	Demonstrates when exploitation is justified.

Handling Uncertainty and Unexpected Results

- Compare predictions with actual outcomes and reassess models when they diverge.
- Use LOOCV, predictive uncertainty and challenger models to assess reliability.
- Distinguish observation noise from model uncertainty where possible, particularly for F3.
- Avoid automatically following acquisition functions when their recommendations are inconsistent with observed evidence.
- Adjust the search region or modelling approach rather than repeatedly pursuing an unsuccessful prediction.

5. **Next steps and reflection**
   - Outline your planned actions or experiments to improve the model's performance.
   - Include two or three sentences connecting your work to the broader ML landscape.
   - Add a short statement on how you'd communicate your results to a non-technical audience or stakeholder.

Planned Actions to Improve Performance

My next step would be to address the specific weaknesses identified in each function and focus on refining the approaches already showing promise rather than introducing additional algorithms.

Specific actions would include:

- **F1 and F2:** Use targeted evaluations to better understand local sensitivity and improve GP predictions around promising regions.
- **F3:** Investigate observation noise through more evaluations. Also, improve GP validation and uncertainty estimates.
- **F4:** Compare GP, NN and TuRBO predictions against results to identify which approach is more reliable.
- **F6 and F7:** Continue refining TuRBO, including trust-region adjustments. Evaluate whether the predictions consistently translate into better outcomes.
- **F5 and F8:** Retain the existing solutions unless further analysis identifies credible opportunities for improvement.

Connection to Broader Machine Learning

This exercise has reinforced the importance of the earlier lesson learned on bias–variance trade-off, model validation and uncertainty estimation. This is particularly true when working with sparse datasets. More complex models do not necessarily generalise better, and predictive accuracy does not always translate into better optimisation decisions.

It has also demonstrated the value of active learning, where the choice of the next observation is part of the modelling process rather than simply collecting more data.

Communicating the Results

For a non-technical stakeholder, I would use an investment analogy in terms of maximising returns from investment decisions with incomplete information and a limited budget. Each evaluation has an opportunity cost. The objective is to decide whether to investigate a new opportunity or concentrate on the current one.